FINAL RESEARCH PROPOSAL

Robust Human-Motion Retrieval from Noisy Pose Sequences

GIACOMO CAPPELLETTO
Computer Engineering / College of Engineering

PROFESSOR BRIAN KULIS
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RESEARCH QUESTION

Given one unmodified example of each new action, does contextual training help a model find the correct match when body joints or video frames are missing or inaccurate?

Project Description and Goals

Video-based pose systems record a person as a time-ordered sequence of estimated body-joint locations. These estimates can jitter, omit joints, or lose frames, causing a model trained on unmodified benchmark data to retrieve the wrong movements when given poses extracted from real video. This project asks whether training that considers groups of nearby examples, rather than only isolated pairs, makes retrieval more reliable when pose data is incomplete or inaccurate.

I will use NTU RGB+D 120, an established dataset containing 120 human actions [3]. Its official one-shot protocol trains on 100 actions and evaluates on 20 unseen actions. One unmodified clip per unseen action forms the reference set; every other eligible clip must be matched to the correct reference. MotionBERT [2] will convert each sequence into a compact numerical representation. I will compare the contextual metric-learning method developed by Liao, Tsiligkaridis, and Kulis [1] with standard contrastive training while keeping the data, encoder, trainable layer, and evaluation procedure fixed.

My primary goal is to determine whether the contextual objective reduces the accuracy loss caused by displaced joints, missing body parts, or missing frames. I hypothesize that it will lose less accuracy as these controlled errors become more severe. I will also report clean-input accuracy, run-to-run variation, computational cost, and common failure cases. I will prepare the data, implement the evaluator and corruptions, adapt the objectives, run the experiments, and analyze the results. Deliverables will include clean-versus-corrupted retrieval curves, documented code and configurations, and a report or poster. The project is limited to matching action classes on one dataset; it will not claim to measure movement quality, diagnose medical conditions, or retrieve directly from raw video.

Project Significance

Searching for motion by example can help researchers organize collections of video-derived poses or motion-capture recordings and support later work in sports analysis, rehabilitation, and human-computer interaction. In these settings, body-joint locations are estimated from images rather than measured perfectly. Viewpoint, occlusion, and estimation errors can make joints shift or disappear, so a method that works only on carefully prepared data may be unreliable on realistic inputs.

Standard metric-learning methods organize numerical representations by comparing individual examples or pairs. Contextual metric learning also considers neighborhoods: two examples should be close when supported by similar nearby examples [1]. It improved image retrieval, but that does not establish robustness to errors in body-joint coordinates over time. Prior pose models address noisy joints through encoder design or data augmentation [7]; this project instead isolates the training objective while holding the motion encoder and evaluation fixed.

The study tests whether this neighborhood-based objective transfers from static images to temporal motion and unseen action classes. A positive result would identify a practical change for more reliable motion search. A null or negative result would show that the image-domain advantage does not automatically transfer and would reveal difficult pose errors. Either outcome will clarify when contextual metric learning is useful and provide a reproducible benchmark. Conclusions will remain limited to action matching on this dataset, not clinical, biomechanical, or identity-related judgments.

Methodology / Process

1. **Build a leakage-resistant benchmark.** I will use MotionBERT's published NTU RGB+D 120 one-shot pipeline and HRNet-derived two-dimensional keypoints [2,3]. The protocol has 100 training actions, 20 novel test actions, and one reference clip per test action. I will reserve a fixed subset of training actions for class-disjoint validation, fix all settings there, retrain on all 100, and evaluate the 20 novel actions only after those choices are locked. Synchronized views of a reference performance will be excluded from its query pool, with automated split and duplicate checks.
2. **Fix the motion representation.** A generic MotionBERT checkpoint not trained on NTU action labels will encode each sequence. I will freeze the encoder and train only an identically initialized, lightweight output layer for each objective. This isolates the objective and is feasible on one GPU. Because MotionBERT was pretrained with masked and noisy poses, I will interpret only the incremental differences between the matched pipelines.
3. **Train matched comparisons.** I will compare contrastive-only training with the published contextual-plus-contrastive objective [1,8]; supervised contrastive learning [4] will be a secondary baseline. All systems will use the same output size, class-balanced batches, optimization method, training duration, three paired seeds, and small tuning budget. Settings will be selected only on unmodified validation data, following fair metric-learning practice [5].
4. **Run controlled stress tests.** Training data and references will remain unmodified. I will alter only test queries at three preset severities using body-size-scaled coordinate jitter, masking of joints or limbs, and removal of consecutive frames. Every method will receive the same altered queries. I will measure top-1 accuracy (whether the first match is correct), mean reciprocal rank (how highly the correct reference ranks), and top-5 recall (whether it appears among the first five). The primary result will be the absolute top-1 accuracy loss and the full accuracy-versus-severity curve, preventing a weak clean model from appearing robust. I will report variation across seeds and paired confidence intervals over the same queries.
5. **Document limits and reproducibility.** I will preserve configurations, split lists, corruption seeds, tests, and resource measurements. The project uses licensed pose data, collects no new participant data, infers no identity or demographic attributes, and follows redistribution rules. I will confirm the appropriate BU human-subjects determination with my mentor before research begins.

Timeline

PERIOD	FOCUS AND OUTPUT
Weeks 1-2	**Data and protocol:** obtain approved NTU files, compatible poses, official class split, and reference list; freeze validation and test episodes; add leakage checks.
Weeks 3-4	**Evaluator and stress tests:** validate feature extraction and clean retrieval; implement metrics and deterministic joint-jitter, joint-masking, and frame-removal tests.
Weeks 5-7	**Matched training:** train the three output layers with identical controls; select settings only on unmodified validation data and resolve unstable runs.
Weeks 8-9	**Repeated evaluation:** run three paired seeds and all prespecified corruption severities; record clean and corrupted accuracy, time, memory, and failed runs.
Weeks 10-11	**Analysis:** calculate accuracy losses and confidence intervals, inspect errors, compare corruption types, and run only essential ablations.
Weeks 12-13	**Finalization:** reproduce figures from saved configurations, document permitted code, and prepare the final report and UROP presentation or poster.

The required result is the one-dataset, one-shot, three-objective comparison. If full feature extraction exceeds available resources, I will use a fixed subset of the training actions. Encoder fine-tuning, another dataset, and real pose-estimator shifts are future work.

Background Experience

I am a Boston University Computer Engineering student with experience in software and machine-learning data pipelines. My coursework and projects have trained me in programming, algorithms, linear algebra, experimental design, and quantitative analysis.

Most directly, I developed a rowing-biomechanics pipeline that processes high-frame-rate video, detects two-dimensional joints with MMPose, lifts them into three dimensions with MotionBERT, and calculates joint trajectories and angles. This work required handling time-series pose data, coordinate normalization, noisy detections, model integration, and numerical validation. It showed me how small joint-estimation errors affect later movement analysis. In a recent software and AI engineering internship, I also built and tested data-intensive applications, strengthening my ability to translate requirements into reliable software.

For this proposal, I reviewed the contextual metric-learning paper and its public implementation [1], studied MotionBERT and pose-retrieval literature [2,6], and completed a focused implementation review. I discussed the question, controls, and feasible scope with Professor Brian Kulis; his expertise in metric learning and retrieval will guide the design and interpretation.

I will prepare and verify the data protocol, implement the evaluator and controlled errors, adapt and test the objectives, run the experiments, analyze uncertainty and failures, and produce the code and final report. My experience as a Division I rower motivates my interest in movement analysis, while this proposal remains a general machine-learning study using an established action dataset.

Bibliography

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